In the present study (Table 1) showed the bioaccumulation of Zinc in gills and muscles of *Clarias batrachus* as 2.69 ± 0.8 µg/g and 1.4± 0.5 µg/g respectively. *Labeo rohita* showed lesser bioaccumulation of Zn in gills and muscles as 1.8±0.6 µg/g and 0.8±0.04 µg/g respectively. Zn concentration in muscles of *C. batrachus* was significantly higher than that of *L. rohita* at 5% significance level. This was in accordance with Yousafzai *et al*. (2017)[4] where they obtained higher concentration for Zn was found in gills as compared to muscles .

Table 2 showed higher bioaccumulation of Copper in *Clarias batrachus* as compared to *Labeo rohita*. Cu was found to be higher in gills of *C. batrachus* as 1.28 ±0.15 µg/g and muscles as 0.2 ±0.08 µg/g. Gills and muscles of *Labeo rohita* showed comparative lesser bioaccumulation of copper as 0.48±0.09 µg/g and 0.08±0.008 µg/g respectively. Statistical analysis showed that bioaccumulation of Cu in gills of *C. batrachus* was significantly higher than that of *L. rohita* at 0.1 % level. Also, bioaccumulation of Cu in Muscles of *C.batrachus* was significantly higher than that of *L. rohita* at 5% significance level. This is different from the findings of Pandey *et al.* (2014)[5] where the high accumulation of Cu was detected in the muscles. From the data, the storage of Cu was highest in the liver in the studied species which is in congruency with Oldewage and Marx (2000)[6] who used *Clarias gariepinus* for their study and obtained that liver had the highest Cu concentration.

According to Forstner and Prosi (1979)[7] the harmful effects of heavy metals as pollutants results from incomplete biological degradation[8]. The water bodies are contaminated by industrial discharge and agricultural wastewater discharge, possible sources of heavy metals, which are absorbed by the suspended sediments which gets precipitated to be a part of the surface sediment, which in case of certain disturbances gets released into the water column and serve as the potential secondary source of contaminants which can be a potential threat to ecosystems.[9]

Fishes are integral component of human diet,[10] because it's a good source of high-quality protein, amino acid (lysine, sulphur containing amino acids etc), iodine, calcium, trace elements (Mn, Fe, Cu, Zn, Se etc), vitamins and omega-3-polyunsaturated fatty acid.[11] Fish are the organisms that absorb dissolved or available metals[12] from the surrounding environment and therefore are considered as an excellent biomarker of metal pollution in an aquatic ecosystem. In an aquatic ecosystem fishes occupy higher tropic levels this is another reason for considering it as a biomarker of metal pollution. Since fishes are among the top consumers in an aquatic environment, the tropic transfer which is also referred to trophodynamics of elements along a food chain can result in an increase (biomagnification) or decrease (bio-dilution) or even no change in the element concentrations from lower to upper components of the food web.[13] Therefore, since they play an important role in human nutrition, fish need to be screened to ensure that unnecessarily high levels of the toxic heavy metals are not being transferred to man through them.

CONCLUSION

Both Zinc and Copper heavy metals were found to be heavily bioaccumulated in the gills and muscles of the two commonly consumed freshwater fishes of Bara Talab i.e., *Clarias batrachus* and *Labeo rohita*.

Such extent of bioaccumulation in both fish species is suspected to be because of the pollutants in the Bara Talab in response to anthropogenic activities taking place in and around it.

Acknowledging it as one of the rice source of edible fish supply in Jharkhand, the Government of Jharkhand should take major steps to protect this fresh water body from any kind of pollution, so as to better the health condition of fishes present there and thereby providing a healthier source of protein to the markets.

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An experimental investigation on growth rate of water lettuce (*Pistia stratiotes* L.)

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Abstract- *Pistia stratiotes* commonly called water lettuce is free floating aquatic herbs is the member of family areaceae. This is an alien invasive species. This grows very fast and soon covers the entire surface of the water bodies. It is often grown as an ornamental plant in the water bodies or small water tubs owing to its beautiful foliage. But due to its rapid growth it causes it causes nuisance to water bodies as because it is very difficult to control once it cover the entire surface. During the course of study of aquatic and semi aquatic angiosperms of Jharkhand it has been observed that several ponds have been covered with the mat of *Pistia stratiotes*. In order to know its pace of spread, an experiment was conducted at department of botany KKM College, Pakur to observe the multiplication. It was initiated in a water tub. Soon the entire surface of the tub was covered with *Pistia*. In the present communication number in increase of *Pistia* has been recorded on daily basis for a month and same has been discussed.

Key words: *Pistia stratiotes,* ornamental plant, nuisance, water bodies.

INTRODUCTION

Pistia, a genus of family araceae is an aquatic floating herb. Commonly called water lettuce or water cabbage. The genus "*Pistia*" is derived from the Greek word "pistos" meaning watery. The specific epithet "*stratiotes*" is derived from a Greek word that means soldier, a possible reference to the species *stratiotes* which contains an aquatic plant species with leaves shaped like swords. It is the single genus of the tribe Pistieae that shows its systematic isolation inside the family. It has only one species i.e., *Pistia stratiotes*, This was first reported from the Nile close to Lake Victoria within Africa. Which further spreaded through naturally or by anthropogenic activities to almost entire tropical and subtropical fresh waterway. Water lettuce is among the world's most dynamic freshwater aquatic plants due to the reason it is considered as highly invasive species. The invasion of this species in the environment leads to serious environmental as well as socio-economic damage to its surrounding the communities. In water with high nutrient load this plants shows luxuriant growth and presents weedy appearance.

Water lettuce is among the world's most dynamic freshwater aquatic plants and is considered as an invasive species. The species can be introduced to new areas by water dispersal, fragmentation, and hitchhiking on marine transportation or fishing equipment.

It reproduces rapidly through vegetative fragmentation with the help of stolon as well as seeds. Under optimal environmental condition in doubles its number in three week. It occurs in the surface of lakes

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and ponds. The heavy growth of this plant hampers the effective uses of water causes clogging of canals and dominates in wetland environments.

Description of the plant:

Free floating herbs having 2-10 cm long leaves, surrounded at its base by a membranous sheath, pubescent, obovate-cuneate; apex rounded and truncate. Spathes 10-12 mm long, yellowish unite, obliquely campanulate, contracted near the middle. Carpels obliquely adnate to the spadix; apex free. Ovary one loculed with numerous ovules on a marginal placentae. It flowers and fruit in the month of August - April.

Systematic position

Kingdom	-	Plantae
Subkingdom	-	Viridiplantae
Infrakingdom	-	Streptophyta
Superdivision	-	Embryophyta
Division	-	Tracheophyta
Subdivision	-	Spermatophytina
Class	-	Magnoliopsida
Superorder	-	Lilianae
Order	-	Alismatales
Family	-	Araceae
Genus	-	*Pistia* L.
Species	-	*P. stratiotes*

MATERIALS & METHOD

One pair of *Pistia stratiotes* was taken and is allowed to grow and multiply in a controlled condition. The increase in number of *Pistia* has been observed on daily basis. The date wise increase in numbers were recorded and photographs of the increased number were taken. On the basis of the same a graph has been plotted.

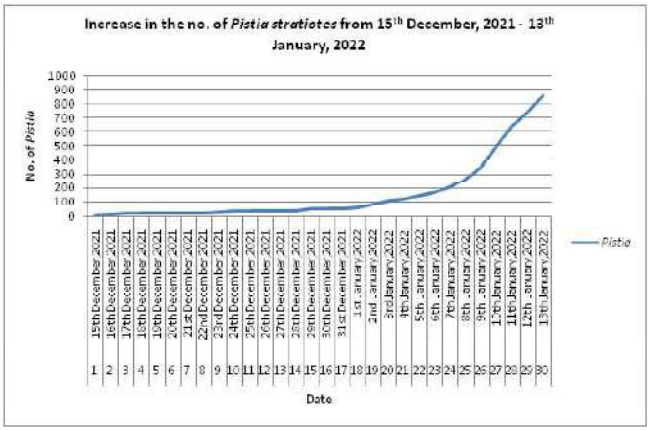
RESULT & DISCUSSION

One pair of *Pistia* was allowed to grow on a controlled condition in a pot on 15th of December, 2021, very next day i.e. on 16th of December, 2021, the number increased gradually to 11, 18, 18, 18, 18, and then 19, 19, 27, 31, 31, 32, 32, 32 further 46, 46, 49, 54, 80, 100, 120, 140, 165, 200, 255, 300, 500, 640,746 and finally 859 on 13th January, 2022.

This plant being alien invasive soon covers the entire surface, causing disturbance in reaching the sunlight for bottom feeder and alters the grazing the food chain. The huge decomposition of *Pistia* settles in the bottom and reduces the depth.

Table 1- Datewise multiplication of *Pistia stratiotes*

Sl. No.	Date	No. of *Pistia*
1.	15th December, 2021	2
2.	16th December, 2021	11
3.	17th December, 2021	18
4.	18th December, 2021	18
5.	19th December, 2021	18
6.	20th December, 2021	18
7.	21st December, 2021	19
8.	22nd December, 2021	19
9.	23rd December, 2021	27
10.	24th December, 2021	31
11.	25th December, 2021	31
12.	26th December, 2021	32
13.	27th December, 2021	32
14.	28th December, 2021	32
15.	29th December, 2021	46
16.	30th December, 2021	46
17.	31st December, 2021	49
18.	1st January, 2022	54
19.	2nd January, 2022	80
20.	3rd January, 2022	100
21.	4th January, 2022	120
22.	5th January, 2022	140
23.	6th January, 2022	165
24.	7th January, 2022	200
25.	8th January, 2022	255
26.	9th January, 2022	350
27.	10th January, 2022	500
28.	11th January, 2022	640
29.	12th January, 2022	746
30.	13th January, 2022	859



Graph 1- Showing the growth pattern of *Pistia* for one month